BOTULINUM TOXIN IN RAYNAUD'S PHENOMENON: A SINGLE CENTER DOUBLE-BLINDED RANDOMIZED CONTROLLED TRIAL

Presenter: Paul A. Ghareeb, MD

Co-Authors: Timothy J. Westbrooks, MD[2]; Victoria A. Conn, BS[2]; Carlton R. Homan, MD[1]; Ciara J. Brown, MD[3]; Owen H. Brown, MD[1]; Krishna N. Chopra, MA[2]; Jaden C. Hardrick, BS[2]; Eric R. Wagner, MD, MS[2]; Michael B. Gottschalk, MD[2]

Affiliations: [1]Emory University Division of Plastic and Reconstructive Surgery, Atlanta, GA; [2]Emory University Department of Orthopedics, Atlanta, GA; [3]MD Anderson Department of Plastic and Reconstructive Surgery, Houston, TX

BACKGROUND: Raynaud's phenomenon (RP) is a vasospastic disorder that can cause significant pain, functional impairment, and digital ischemia. Botulinum toxin type A (BTX-A) has emerged as a potential treatment, but high-quality evidence remains limited. This double-blind, randomized controlled trial evaluated the efficacy of BTX-A injections in improving pain, function, and physiologic perfusion measures in patients with refractory RP.

METHODS: Adults with RP and persistent symptoms despite medical therapy were equally randomized to receive periarterial injections of placebo, 50 units of BTX-A, or 100 units of BTX-A per hand. Participants were assessed at baseline and at four, 12, and 24 weeks post-injection. The primary outcome was change in visual analog scale (VAS) pain. Secondary outcomes included PROMIS Upper Extremity function, QuickDASH, Raynaud's Condition Score, subjective hand value, fingertip temperature, and fingertip oxygen saturation.

RESULTS: Twenty-four subjects were enrolled, with 20 completing 24-week follow-up. There were no significant differences between BTX-A and placebo groups for change in VAS pain at any time point, although a nonsignificant reduction in pain was observed at four weeks post-injection in the pooled BTX-A group (-0.94 ± 0.60; $p=0.13$). By-arm analyses showed similar trends without dose–response effects. Secondary outcomes, including functional scores and physiologic parameters, showed no between-group differences through 24 weeks.

CONCLUSION: BTX-A did not demonstrate statistically significant improvements in pain, function, or digital perfusion compared with placebo among patients with refractory RP. Larger, adequately powered studies with standardized injection techniques and stratified patient groups are needed to clarify the role of BTX-A in Raynaud's management.

A PROPENSITY SCORE-MATCHED ANALYSIS OF BREAST RECONSTRUCTION OUTCOMES STRATIFIED BY TYPE OF ACELLULAR DERMAL MATRIX

Presenter: Annie Glenney

Co-Authors: Naomi Ghahrai, BS[1]; Jace Boswell, BS[2]; Virginia Bailey, BA[1]; Christopher Campbell, MD[1]

Affiliation: [1]Department of Plastic Surgery, Maxillofacial, and Oral Health, University of Virginia Health System, Charlottesville, VA

INTRODUCTION: Acellular dermal matrix (ADM) is commonly incorporated in breast reconstruction to enhance soft tissue support. However, literature comparing ADM types remains limited. Differences in processing, sterility, and biologic integration may contribute to outcomes. This multi-institutional study evaluates complications associated with major ADM types in prosthetic breast reconstruction.

METHODS: The TriNetX US Collaborative Network, a database including over 114 million patients from 67 healthcare organizations between 2007 and November 2025, was queried to identify patients undergoing mastectomy with direct-to-implant or tissue expander placement. Same-day ADM use and specific ADM types were identified using Q-codes. Table 1 lists the five ADM products included, along with descriptors of their processing characteristics. Patients were stratified by ADM use and type. Propensity-score matching balanced baseline characteristics. Post-operative outcomes within 90 days, including implant/TE